Model Reference Adaptive Control Homework Assignment

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Question 1

For the system:

$$W_p(s) = \frac{s+1}{(s-2)(s-1)} = \frac{s+1}{s^2-3s+2}$$

design a Model Reference Adaptive Controller (MRAC) such that the output tracks the reference model:

$$W_m(s) = \frac{1}{s+1}$$

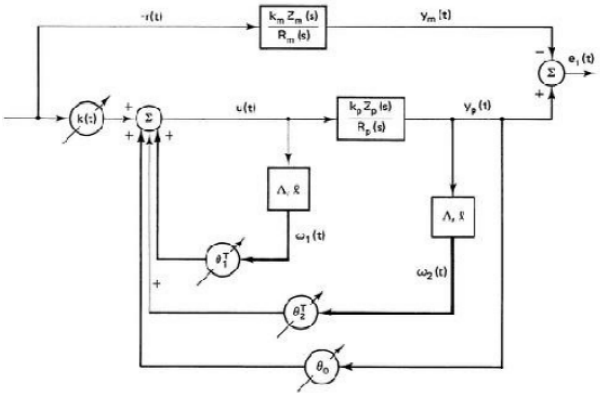


Figure 1: Block diagram of the adaptive control system

Algebraic Solution

Since the reference model has a lower degree than the plant, we multiply it by $\frac{s+1}{s+1}$ to match the degree:

$$W_m(s) = \frac{1}{s+1} \cdot \frac{s+1}{s+1} = \frac{s+1}{(s+1)^2}$$

Now, define the plant transfer function:

$$W_p(s) = \frac{Z_p(s)}{R_p(s)} = \frac{s+1}{s^2-3s+2}, \quad Z_p(s) = s+1, \quad R_p(s) = s^2-3s+2$$

And the modified reference model:

$$W_m(s) = \frac{Z_m(s)}{R_m(s)} = \frac{s+1}{(s+1)^2}, \quad Z_m(s) = s+1, \quad R_m(s) = (s+1)^2$$

The adaptive control law structure leads to the following matching conditions:

$$\begin{aligned} \lambda = Z_m(s) = s+1 \Rightarrow \lambda = 1 \\ \lambda - C^*(s) = Z_p(s) = s+1 \Rightarrow C^*(s) = s+1-(s+1) = 0 \\ R_p(s) - k_p D^*(s) = R_m(s) = (s+1)^2 \Rightarrow D^*(s) = \frac{R_p(s) - R_m(s)}{k_p} \end{aligned}$$

Now compute:

$$D^*(s) = \frac{s^2 - 3s + 2 - (s^2 + 2s + 1)}{k_p} = \frac{-5s+1}{k_p}$$

Let us define:

$$\Lambda = [-1], \quad l = 1 \Rightarrow (sI - \Lambda)^{-1} = \frac{1}{s+1}$$

Now use the canonical parameterization:

$$\frac{C^*(s)}{\lambda(s)} = \theta_1^{*T}(sI - \Lambda)^{-1} \Rightarrow \frac{0}{s+1} = \theta_1^* \cdot \frac{1}{s+1} \Rightarrow \theta_1^* = 0$$

Similarly, for:

$$\frac{D^*(s)}{\lambda(s)} = \theta_0^* + \theta_2^{*T}(sI - \Lambda)^{-1}$$

Use:

$$\frac{-5s+1}{s+1} = -5 + \frac{6}{s+1} \Rightarrow \theta_0^* = -5, \quad \theta_2^* = 6$$

Finally, the adaptive parameters are:

$$\boxed{\theta_0^* = -5, \quad \theta_1^* = 0, \quad \theta_2^* = 6}$$

Analytical Stability Analysis

Since the reference model $W_m(s) = \frac{s+1}{(s+1)^2}$ is a Strictly Positive Real (SPR) transfer function, the Lyapunov stability of the adaptive system can be guaranteed using the Meyer-Kalman-Yakubovich (MKY) Lemma.

Let the output estimation error be defined as:

$$e_1 = y_p - y_m$$

We define the adaptation law based on the SPR framework as follows:

$$\dot{\phi} = \text{sign}(k_p) \cdot e_1 \cdot \omega \quad \text{where} \quad \omega = \begin{bmatrix} r \\ \omega_1 \\ y_p \\ \omega_2 \end{bmatrix}$$

Here, ϕ is the parameter error vector: $\phi = \theta - \theta^*$, and ω is the regressor vector composed of filtered signals and measurable quantities.

The full parameter vector is defined as:

$$\theta = \begin{bmatrix} k \\ \theta_1 \\ \theta_0 \\ \theta_2 \end{bmatrix}$$

Given the SPR nature of the model and the linear adaptation law, the Lyapunov function can be defined to ensure global asymptotic stability of the tracking error. The MKY Lemma ensures that for SPR models and appropriately structured regressor signals, the energy in the estimation error decreases over time.

Therefore, the adaptive system is stable under the proposed law.

Simulation Results (Question 1)

Figure 2 shows the overall Simulink implementation for the adaptive control system of Question 1.

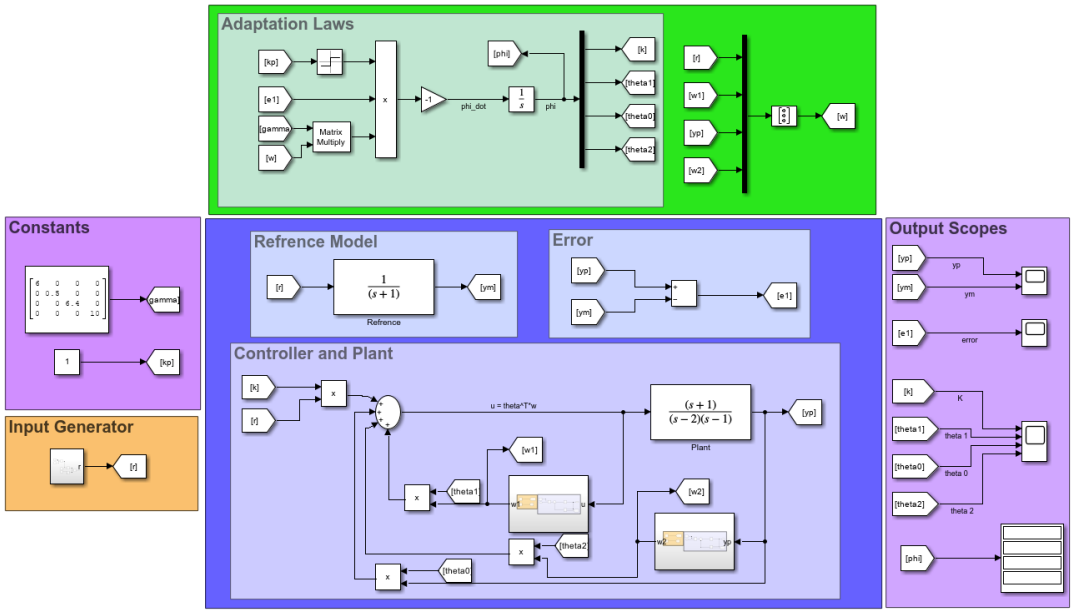


Figure 2: Overall Simulink implementation of MRAC for Question 1

In this section, the simulation results of the adaptive controller for Question 1 are presented. The plant and the reference model are excited by different input signals including step, sinusoidal, and white noise. The tracking error $e_1 = y_p - y_m$ and adaptive parameters are also analyzed.

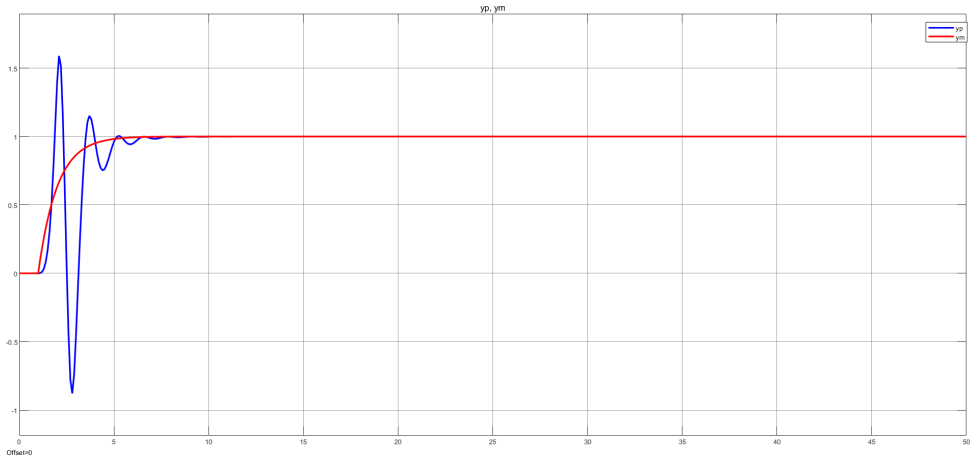


Figure 3: Plant and model output for step input

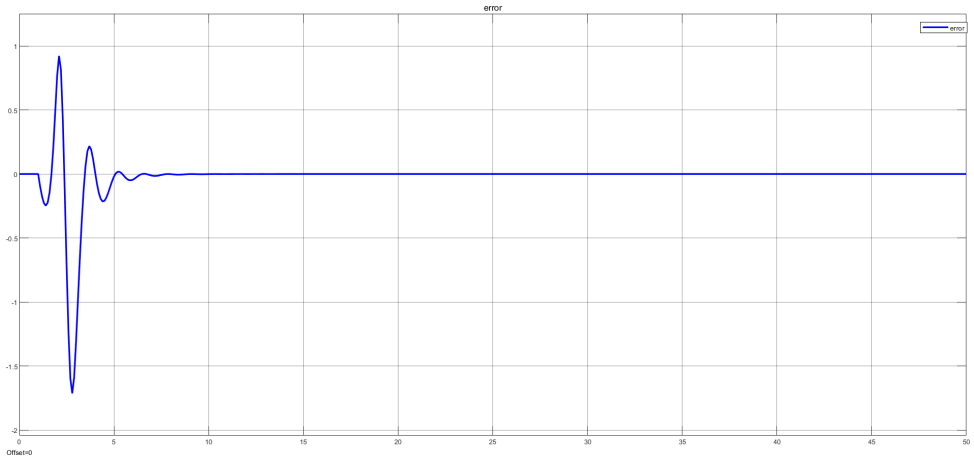


Figure 4: Tracking error $e_1 = y_p - y_m$ for step input

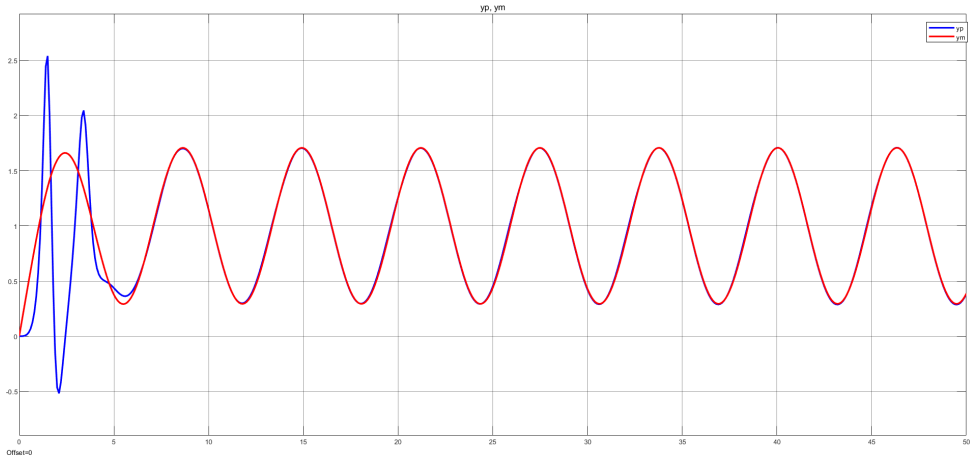


Figure 5: Plant and model output for sinusoidal input ($\omega = 1$)

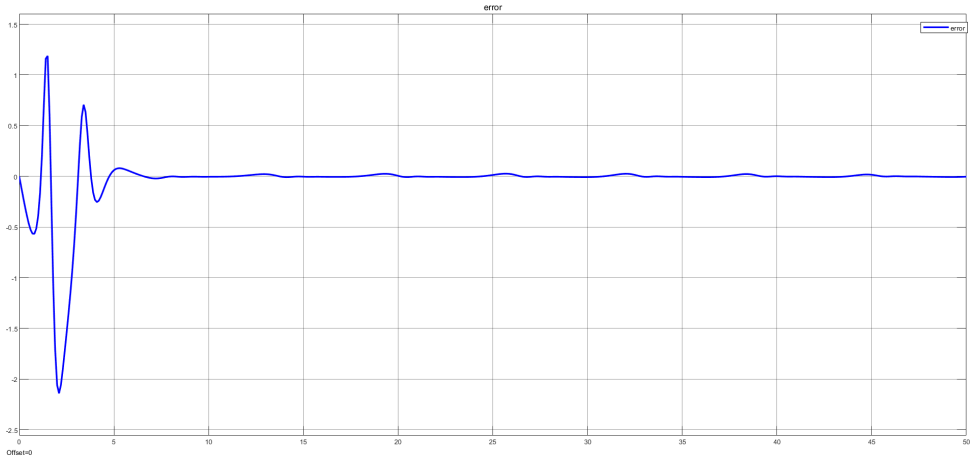


Figure 6: Tracking error e_1 for sinusoidal input ($\omega = 1$)

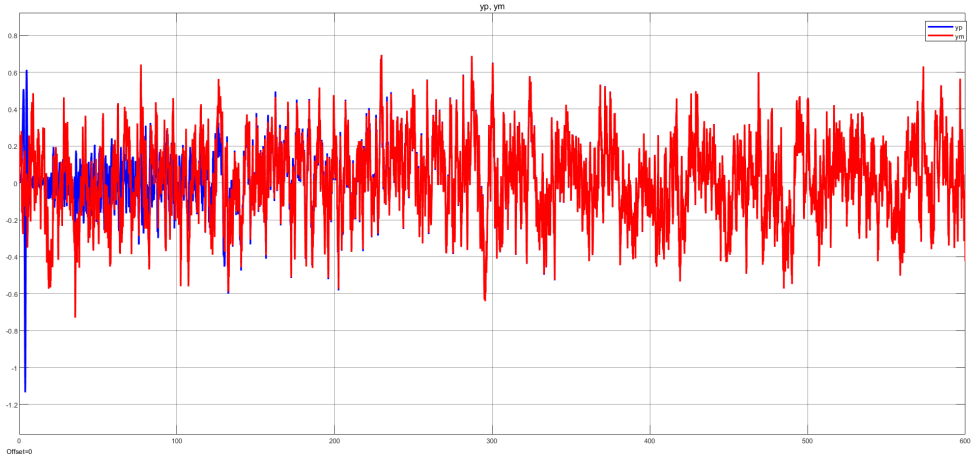


Figure 7: Plant and model output for white noise input

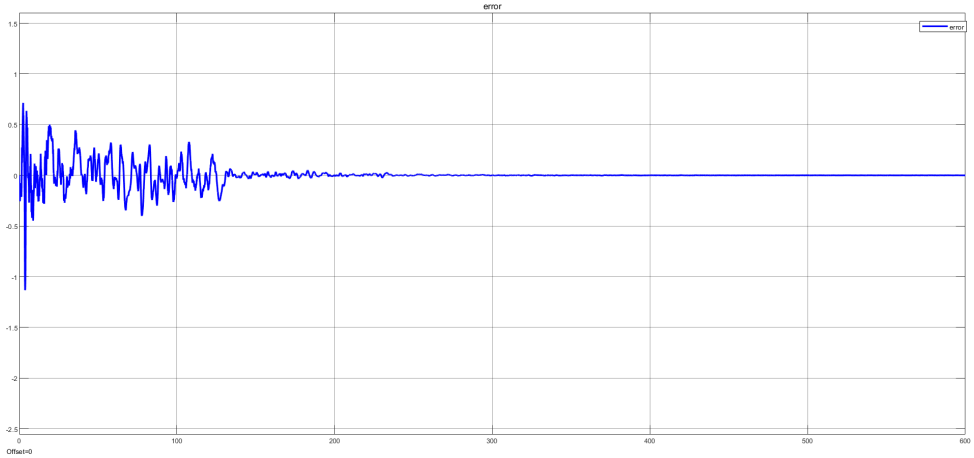


Figure 8: Tracking error e_1 for white noise input

Adaptive Parameter Convergence

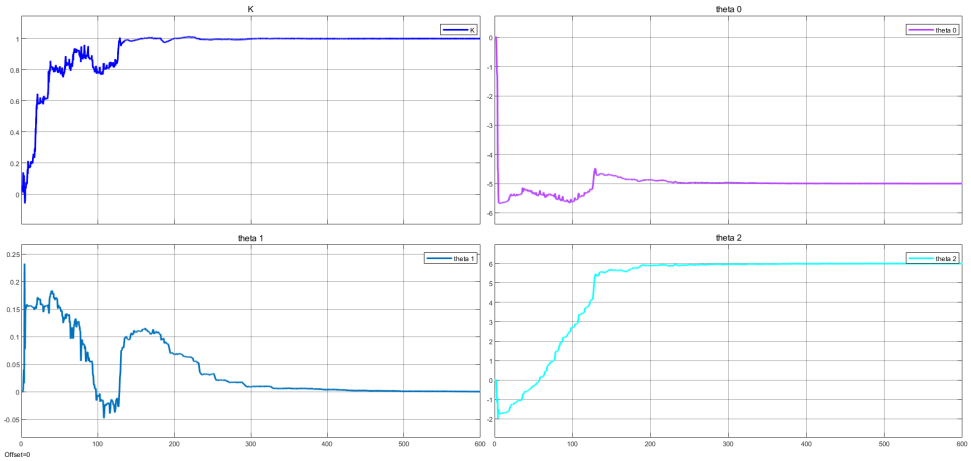


Figure 9: Estimated adaptive parameters under white noise input

Conclusion (Question 1)

The simulation results show that the adaptive controller successfully tracks the reference model under various input types.

In particular, for step and sinusoidal inputs, the tracking error $e_1 = y_p - y_m$ converges with a settling time of less than 10 seconds. However, under white noise excitation, the tracking performance is more sensitive and less ideal due to the high-frequency components. Nevertheless, the adaptive parameters successfully converged to the same values derived from the algebraic solution, validating the theoretical results.

It was observed that systems with fewer poles (lower model order) are easier to track, especially for smoother reference inputs.

To improve convergence speed, a diagonal adaptation gain matrix Γ was selected as follows:

$$\Gamma = \begin{bmatrix} 0.6 & 0 & 0 & 0 \\ 0 & 0.7 & 0 & 0 \\ 0 & 0 & 5.4 & 0 \\ 0 & 0 & 0 & 11 \end{bmatrix}$$

The chosen gain values accelerate the adaptation of corresponding parameters and contribute to the stability and performance of the controller.

Question 2

For the system:

$$W_p(s) = \frac{1}{s(s-1)}$$

design a Model Reference Adaptive Controller such that the output tracks the reference model:

$$W_m(s) = \frac{1}{(s+1)^2}$$

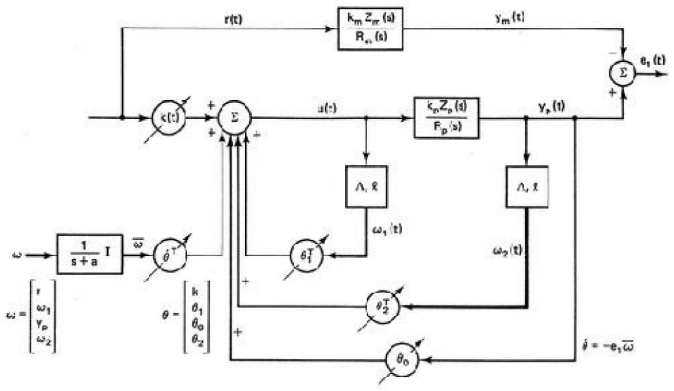


Figure 10: Block diagram of the MRAC system for Question 2

Algebraic Solution

In this section, we solve Question 2 using the algebraic method. The goal is to design both feedforward and feedback controllers such that the overall system transfer function matches the reference model $W_m(s)$.

We use the Bezout identity in the form:

$$MQ - NP = Q^*$$

The system structure defines the following relationships:

$$\frac{D^*(s)}{\lambda(s)} = \theta_0 + \theta_2 \cdot (sI - \Lambda)^{-1} \quad , \quad \frac{C^*(s)}{\lambda(s)} = \theta_1 \cdot (sI - \Lambda)^{-1}$$

To compute $C^*(s)$ and $D^*(s)$, we first reformulate the Bezout equation using:

$$Q = R_p, \quad M = \lambda(s) - C^*(s), \quad P = k_p Z_p(s), \quad N = D^*(s), \quad Q^* = R_m(s) Z_p(s) \lambda_1(s)$$

Where:

- $R_p(s)$: denominator of the plant
- $R_m(s)$: denominator of the reference model
- $Z_p(s)$: numerator of the plant
- $\lambda(s)$: observer characteristic polynomial

- $\lambda_1(s)$: a stable first-order polynomial (e.g., $s+1$)

We now write the correct Bezout equation (with a minus sign):

$$(\lambda(s) - C^*(s))R_p(s) - D^*(s) \cdot k_p Z_p(s) = R_m(s)Z_p(s)\lambda_1(s)$$

Assume:

$$\Lambda = -1, \quad l = 1 \Rightarrow (sI - \Lambda)^{-1} l = \frac{1}{s+1}$$

From this, we get:

$$\frac{D^*(s)}{\lambda(s)} = \theta_0 + \theta_2 \cdot \frac{1}{s+1} \Rightarrow D^*(s) = \theta_0(s+1) + \theta_2$$

$$\frac{C^*(s)}{\lambda(s)} = \theta_1 \cdot \frac{1}{s+1} \Rightarrow C^*(s) = \theta_1$$

Now substitute into the Bezout equation:

$$((s+1) - \theta_1)(s^2 - s) - (\theta_0(s+1) + \theta_2) = (s+1)^3$$

Compute the right-hand side:

$$(s+1)^3 = s^3 + 3s^2 + 3s + 1$$

Now expand the left-hand side:
First:

$$(s+1)(s^2 - s) = s^3 + s^2 - s^2 - s = s^3 - s$$

Then:

$$\begin{aligned}((s+1) - \theta_1)(s^2 - s) &= s^3 - s - \theta_1(s^2 - s) \\ &= s^3 - s - \theta_1 s^2 + \theta_1 s\end{aligned}$$

Now subtract:

$$\theta_0(s+1) + \theta_2 = \theta_0 s + \theta_0 + \theta_2$$

Total left-hand side:

$$s^3 - \theta_1 s^2 + (\theta_1 - 1 - \theta_0)s - (\theta_0 + \theta_2)$$

Now match with:

$$s^3 + 3s^2 + 3s + 1$$

Matching coefficients:

$$\begin{aligned}-\theta_1 = 3 \Rightarrow \theta_1 = -3 \\ \theta_1 - 1 - \theta_0 = 3 \Rightarrow -3 - 1 - \theta_0 = 3 \Rightarrow \theta_0 = -7 \\ -(\theta_0 + \theta_2) = 1 \Rightarrow \theta_0 + \theta_2 = -1 \Rightarrow \theta_2 = 6\end{aligned}$$

Final parameter values:

$$\boxed{\theta_0^* = -7, \quad \theta_1^* = -3, \quad \theta_2^* = 6}$$

Finally, from the gain relation:

$$k^* \cdot k_p = k_m \Rightarrow k^* = \frac{k_m}{k_p}$$

Given:

$$k_p = 1, \quad k_m = 1 \Rightarrow \boxed{k^* = 1}$$

Analytical Stability Analysis

The reference model has a relative degree of $n^* = 2$, while the standard conditions for applying the Meyer-Kalman-Yakubovich (MKY) Lemma require a relative degree of 1.

To satisfy this condition, we multiply the reference model $W_m(s)$ by a stable first-order polynomial $s+\alpha$, and correspondingly modify the input signal to include a filtered version of the parameter derivatives.

This modification leads to an augmented regressor structure as illustrated in the block diagram (Figure 11). The augmented control law becomes:

$$u = \theta^T \omega + \dot{\theta}^T \bar{\omega}$$

where:

- ω: filtered input and output signals
- $\bar{\omega}$: auxiliary filtered regressor
- θ: adaptive parameter vector

The adaptation law is chosen as:

$$\dot{\phi} = -\operatorname{sign}(k_p) \cdot e_1 \cdot L \cdot \bar{\omega}$$

where $\phi = \hat{\theta} - \theta^*$ is the parameter estimation error, and L is a stable filter (e.g., $\frac{1}{s+\alpha}$).

Given that the modified model is SPR (Strictly Positive Real) and the regressor structure is properly augmented, we can apply the MKY Lemma to show that the Lyapunov function decreases over time. Therefore, the tracking error $e_1 = y_p - y_m$ and parameter estimation error ϕ are globally bounded, and the adaptive system is stable.

Simulation Results (Question 2)

The following figures demonstrate the simulation results of the adaptive controller designed for Question 2. The system is tested under various input signals including step, sinusoidal, and white noise. The figures show both model and plant outputs, the tracking error $e_1 = y_p - y_m$, and the evolution of adaptive parameters.

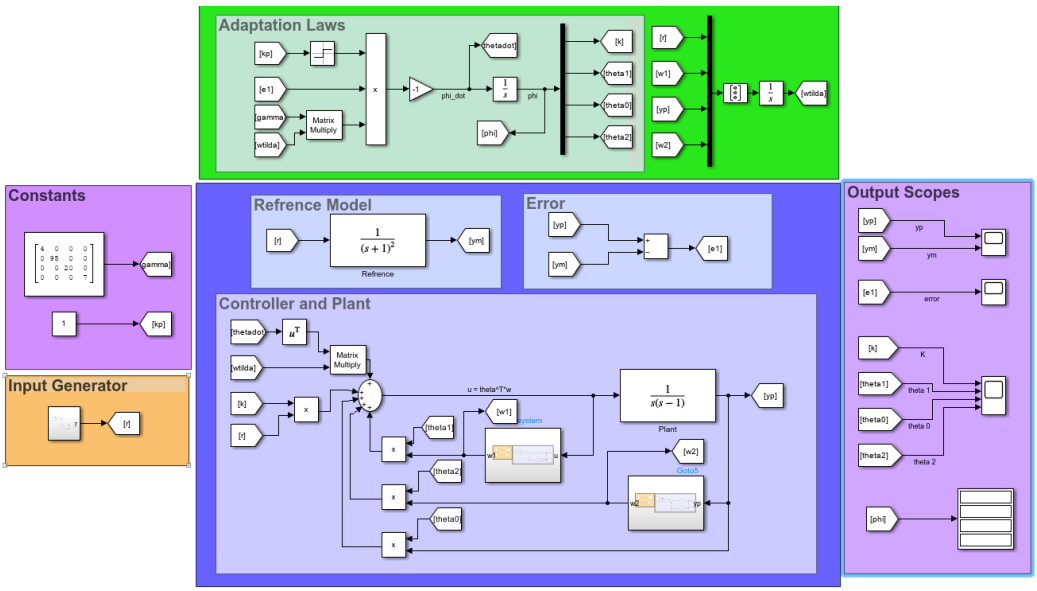


Figure 11: Overall Simulink implementation of MRAC system for Question 2

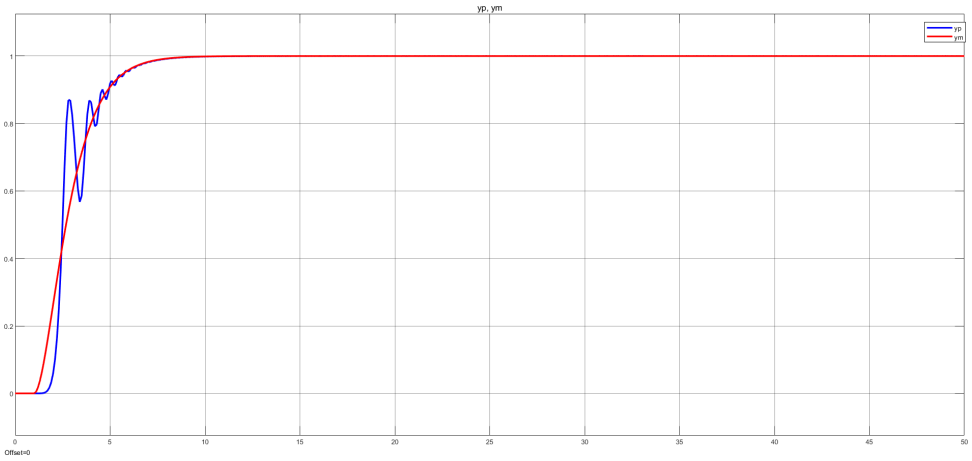


Figure 12: Plant and model output for step input (Q2)

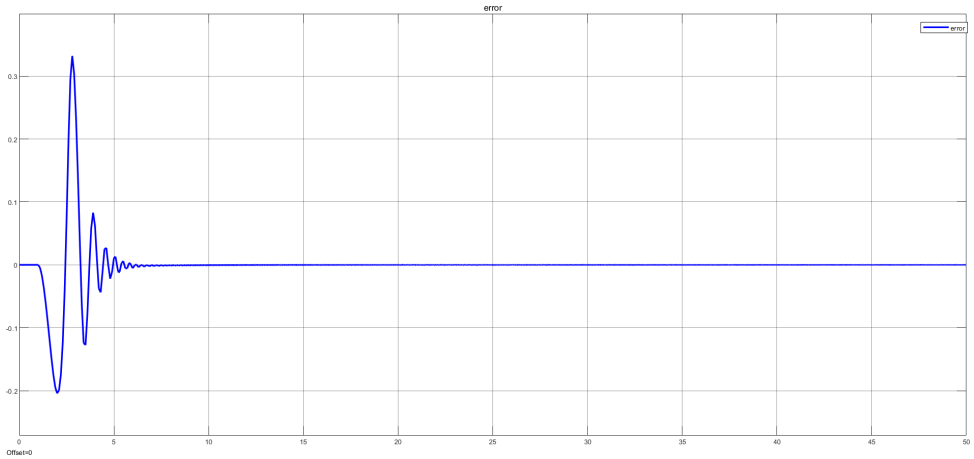


Figure 13: Tracking error e_1 for step input (Q2)

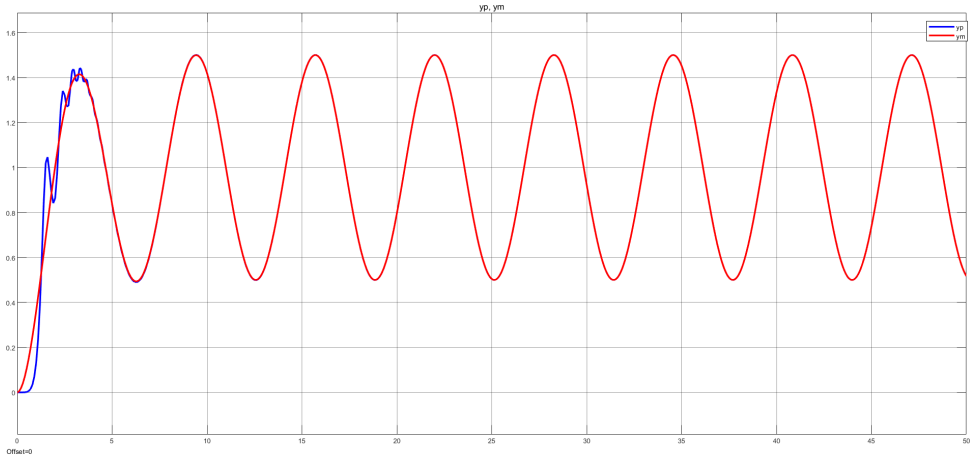


Figure 14: Plant and model output for sinusoidal input ($\omega = 1$) (Q2)

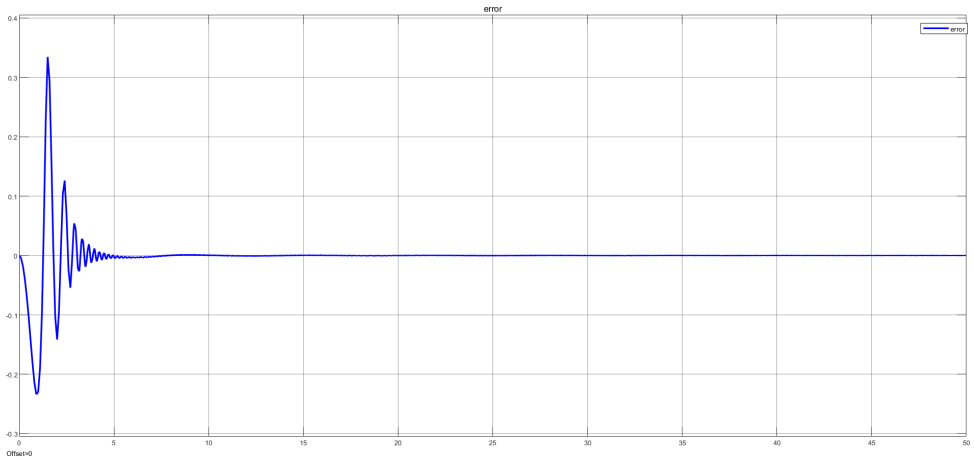


Figure 15: Tracking error e_1 for sinusoidal input (Q2)

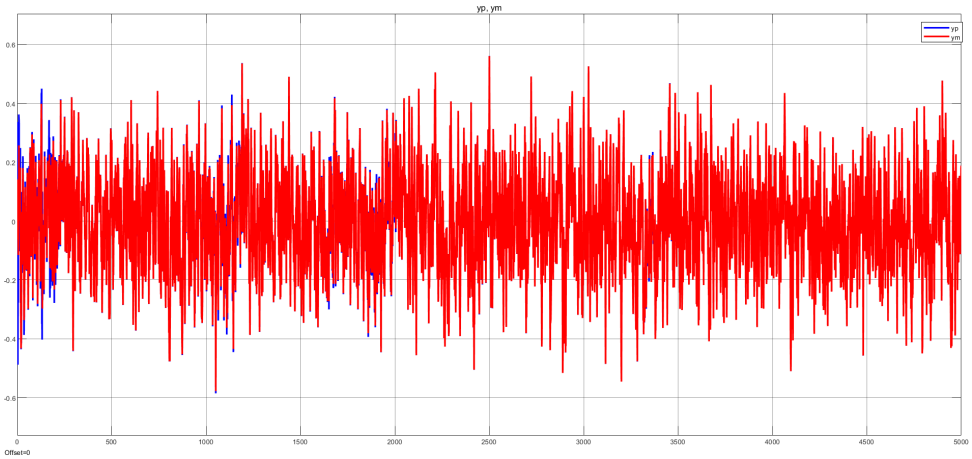


Figure 16: Plant and model output for white noise input (Q2)

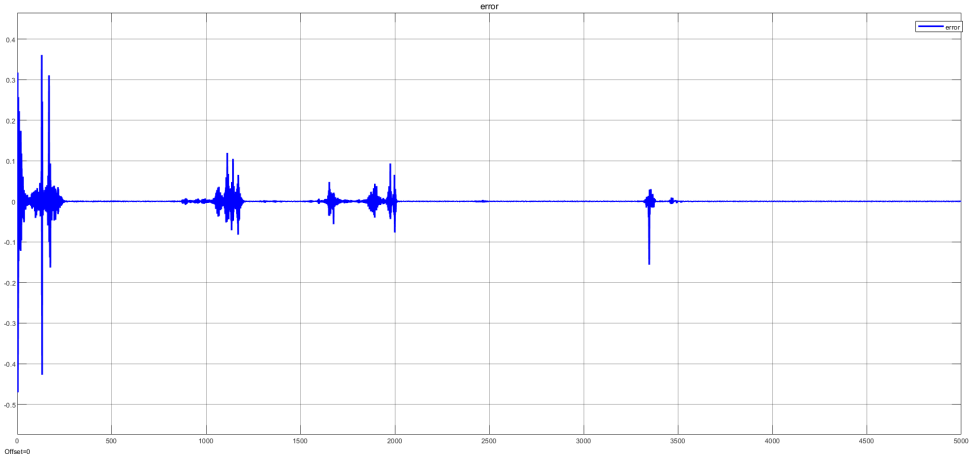


Figure 17: Tracking error e_1 for white noise input (Q2)

Adaptive Parameter Convergence

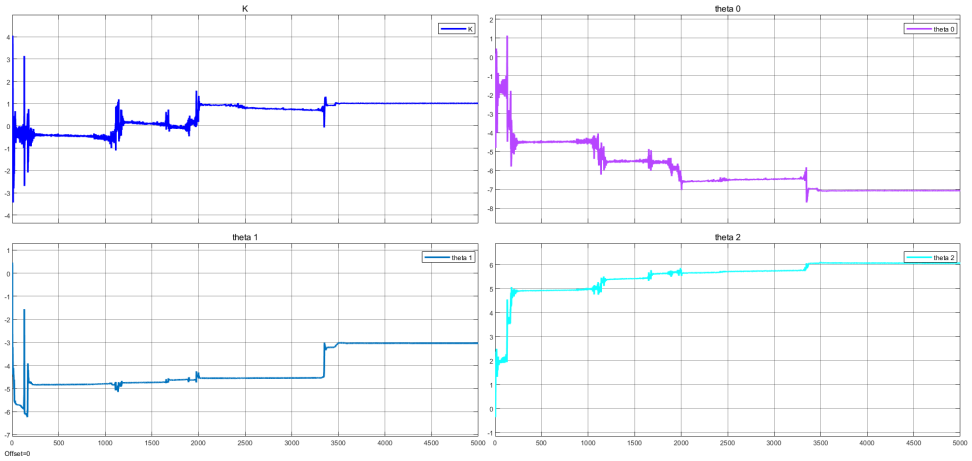


Figure 18: Adaptive parameter convergence under white noise input (Q2)

Conclusion (Question 2)

The simulation results confirm the effectiveness of the adaptive controller designed for Question 2. As in Question 1, the plant successfully tracks the reference model under various input signals such as step, sinusoidal, and white noise.

In both step and sinusoidal inputs, the system demonstrates fast convergence with acceptable settling time. Under white noise input, despite increased fluctuations in tracking error due to the high-frequency nature of the excitation, the system remains stable.

Importantly, the adaptive parameters converge to the same values obtained from the algebraic solution, validating the controller design.

To accelerate parameter adaptation and improve performance, the following diagonal adaptation gain matrix Γ was used:

$$\Gamma = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 95 & 0 & 0 \\ 0 & 0 & 20 & 0 \\ 0 & 0 & 0 & 7 \end{bmatrix}$$

This choice ensures faster convergence of key parameters while maintaining system stability and robustness.

Question 3

For the system:

$$W_p(s) = \frac{1}{(s+1)(s+2)(s+3)}$$

design a Model Reference Adaptive Controller to track the reference model:

$$W_m(s) = \frac{64}{(s+4)^3}$$

using improved adaptive laws (Type-1 and Type-2 error augmentation).

Algebraic Solution Using the Bezout Equation (Question 3)

In this question, the relative degree of the system is $n^* = 3$, so we must augment the reference model with a second-order stable polynomial $\lambda_1(s)$ to satisfy the matching condition for adaptive control.

We start from the general Bezout identity:

$$(\lambda(s) - C^*(s))R_p(s) - D^*(s) \cdot k_p Z_p(s) = R_m(s) \cdot Z_p(s) \cdot \lambda_1(s)$$

Given:

- $R_p(s) = s^3 + 6s^2 + 11s + 6$
- $R_m(s) = s^3 + 12s^2 + 48s + 64$
- $Z_p(s) = Z_m(s) = 1$, $k_p = 1$, $k_m = 64 \Rightarrow k^* = 64$
- $\Lambda = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$, $l = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Step 1: Computing $\lambda(s)$
We compute the denominator of the regressor filter:

$$(sI - \Lambda) = \begin{bmatrix} s & -1 \\ 1 & s+1 \end{bmatrix} \Rightarrow \det(sI - \Lambda) = s^2 + s + 1 \Rightarrow \boxed{\lambda(s) = s^2 + s + 1}$$

Step 2: Parameterization of θ_1 and θ_2
Since $(sI - \Lambda)^{-1} l \in \mathbb{R}^{2\times 1}$, both θ_1 and θ_2 are 2×1 vectors:

$$\theta_1 = \begin{bmatrix} \theta_{11} \\ \theta_{12} \end{bmatrix}, \quad \theta_2 = \begin{bmatrix} \theta_{21} \\ \theta_{22} \end{bmatrix}$$

Step 3: Constructing controller polynomials

$$C^*(s) = \theta_1^T (sI - \Lambda)^{-1} l \cdot \lambda(s), \quad D^*(s) = \theta_0 \lambda(s) + \theta_2^T (sI - \Lambda)^{-1} l$$

Assume:

$$(sI - \Lambda)^{-1} l = \frac{1}{\lambda(s)} \cdot \begin{bmatrix} s+1 \\ 2s+1 \end{bmatrix} \Rightarrow C^*(s) = \theta_{11}(s+1) + \theta_{12}(2s+1)$$

$$D^*(s) = \theta_0(s^2 + s + 1) + \theta_{21}(s + 1) + \theta_{22}(2s + 1)$$

Step 4: Bezout identity substitution

We substitute all expressions into the Bezout identity:

$$(\lambda(s) - C^*(s)) \cdot R_p(s) - D^*(s) = R_m(s) \cdot \lambda(s)$$

Using:

$$\lambda(s) = s^2 + s + 1$$

And previously defined expressions:

$$C^*(s) = \theta_{11}(s + 1) + \theta_{12}(2s + 1), \quad D^*(s) = \theta_0 \cdot (s^2 + s + 1) + \theta_{21}(s + 1) + \theta_{22}(2s + 1)$$

We write:

$$((s^2 + s + 1) - (\theta_{11}(s + 1) + \theta_{12}(2s + 1))) \cdot (s^3 + 6s^2 + 11s + 6) -$$
$$\Big[\theta_0(s^2 + s + 1) + \theta_{21}(s + 1) + \theta_{22}(2s + 1)\Big] = (s^3 + 12s^2 + 48s + 64)(s^2 + s + 1)$$

Now expand both sides completely and collect like terms in powers of s. Then equate coefficients from both sides (for s^5 to s^0) to obtain a system of equations in the unknowns:

Solving the system yields the following results:

$$\boxed{\theta_0^* = 7, \quad \theta_{11}^* = -8, \quad \theta_{12}^* = 1, \quad \theta_{21}^* = -57, \quad \theta_{22}^* = 34}$$

Analytical Stability Analysis Using Error Model 4

In this case, since the relative degree $n^* = 3$ and the plant does not yield a strictly positive real (SPR) error model, we employ Augmented error as discussed in Class.

Figure 19 illustrates the block diagram of this approach.

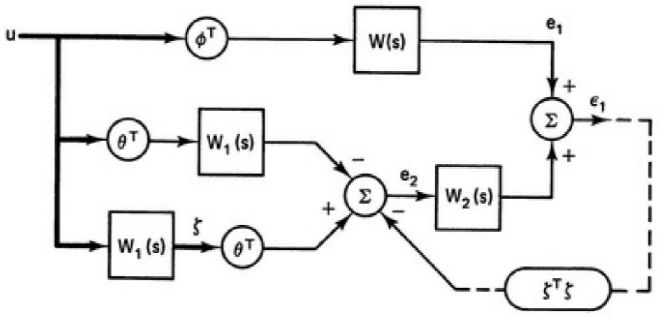


Figure 19: Error Model 4 structure for adaptive control (Method 1)

We factor the model as:

$$W_m(s) = W_1(s) \cdot W_2(s), \quad W_1(s) = \frac{1}{(s+4)^2}, \quad W_2(s) = \frac{64}{s+4}$$

where $W_2(s)$ is SPR, enabling use of MKY Lemma.
We define the following:
- Regressor vector:

$$\omega = \begin{bmatrix} r \\ \omega_1 \\ y_p \\ \omega_2 \end{bmatrix}, \quad \bar{\omega} = \begin{bmatrix} \omega_1 \\ y_p \\ \omega_2 \end{bmatrix}$$

- Filtered signals:

$$\xi = W_1(s) \cdot I \cdot \omega, \quad \bar{\xi} = W_1(s) \cdot I \cdot \bar{\omega}$$

- Auxiliary error:

$$e_2(t) = \left[\theta^T W_1(s) I - W_1(s)\theta^T\right] \bar{\omega}$$

- Augmented error:

$$\epsilon_1(t) = e_1(t) + W_2(s)e_2(t)$$

Using this structure, the adaptive laws are:
- For parameter vector $\phi = \theta - \theta^*$:

$$\dot{\phi}(t) = -\text{sign}(k_p) \cdot \epsilon_1(t) \cdot \xi(t)$$

- For adaptive gain $\psi = k - k^*$:

$$\dot{\psi}(t) = -\epsilon_1(t) \cdot e_2(t)$$

These equations yield a closed-loop error structure similar to Error Model 3, enabling Lyapunov-based analysis using the MKY Lemma.

Given that $W_2(s)$ is SPR and all regressor signals are bounded, the augmented error $\epsilon_1 \in L_2 \cap L_\infty$, and the following properties hold:

- $\epsilon_1(t), e_2(t) \to 0$ as $t \to \infty$
- $\tilde{\theta}(t), \tilde{k}(t)$ remain bounded and converge asymptotically
- Tracking error $e_1(t) \to 0$

Thus, global asymptotic stability of the adaptive system is guaranteed using this structure.

Simulation Results (Question 3)

The following figures illustrate the simulation outcomes of the adaptive controller for Question 3. The system is tested under step and sinusoidal input signals. The outputs of the plant and reference model, the tracking error e_1, auxiliary error e_2, and augmented error ϵ_1 are shown.

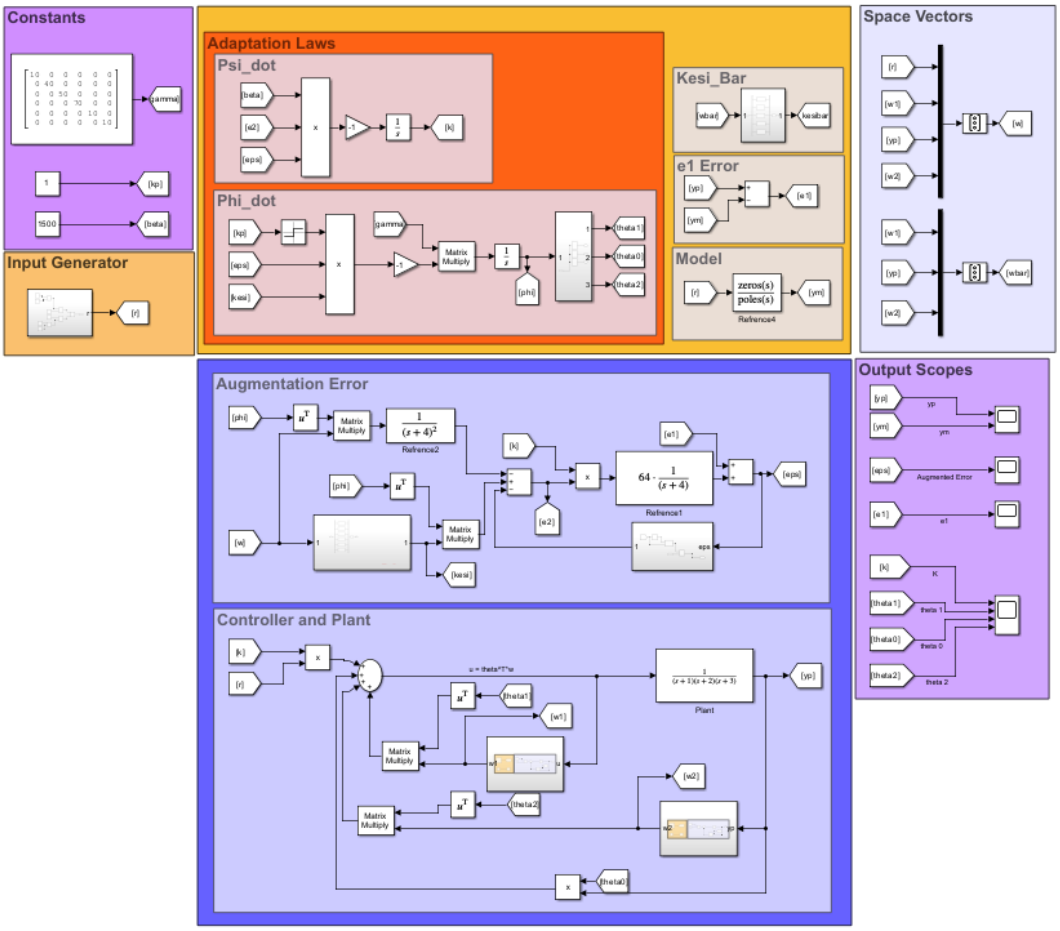


Figure 20: Simulink diagram of the adaptive control system for Question 3

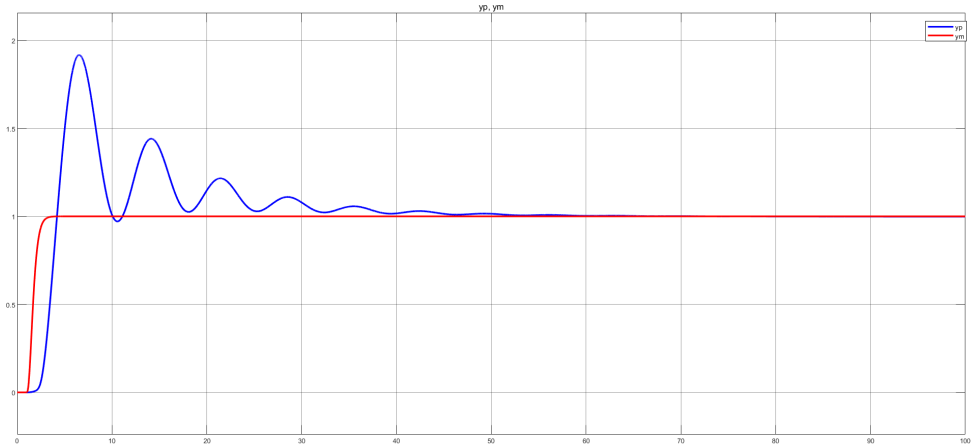


Figure 21: Plant and model output for step input (Q3)

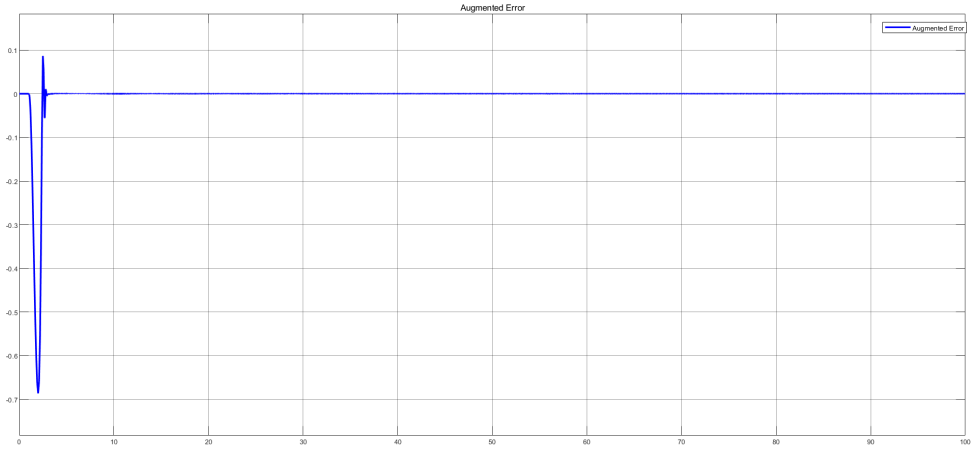


Figure 22: Augmented error ϵ_1 under step input

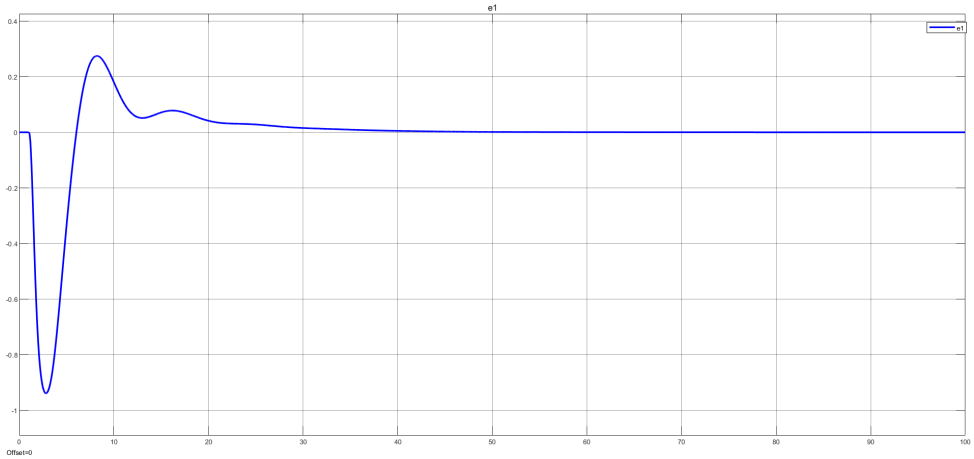


Figure 23: Tracking error e_1 under step input

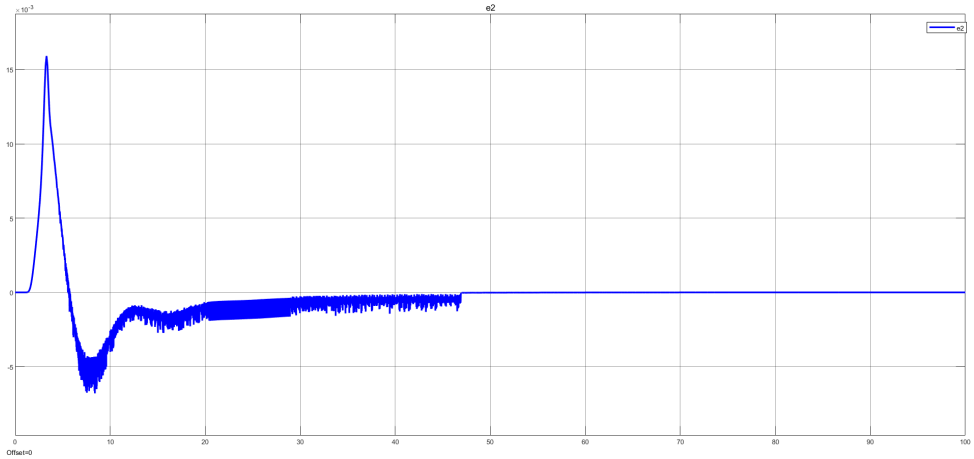


Figure 24: Auxiliary error e_2 under step input

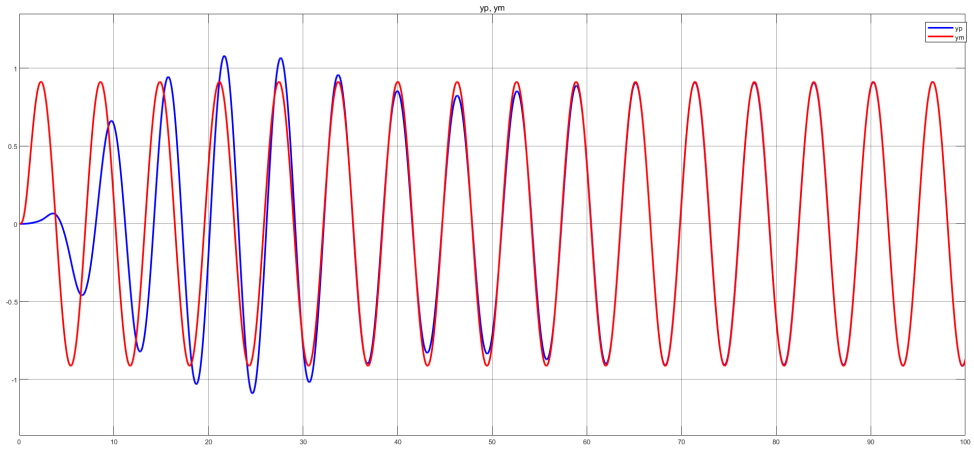


Figure 25: Plant and model output for sinusoidal input (Q3)

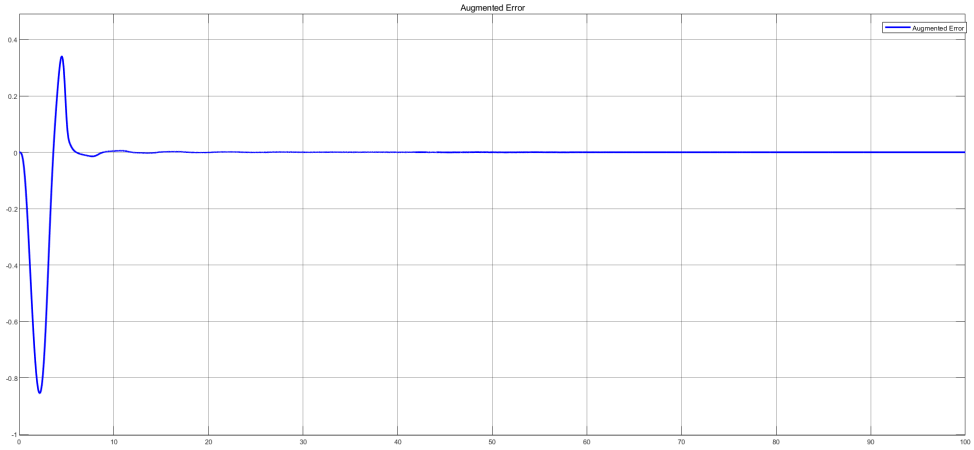


Figure 26: Augmented error ϵ_1 under sinusoidal input

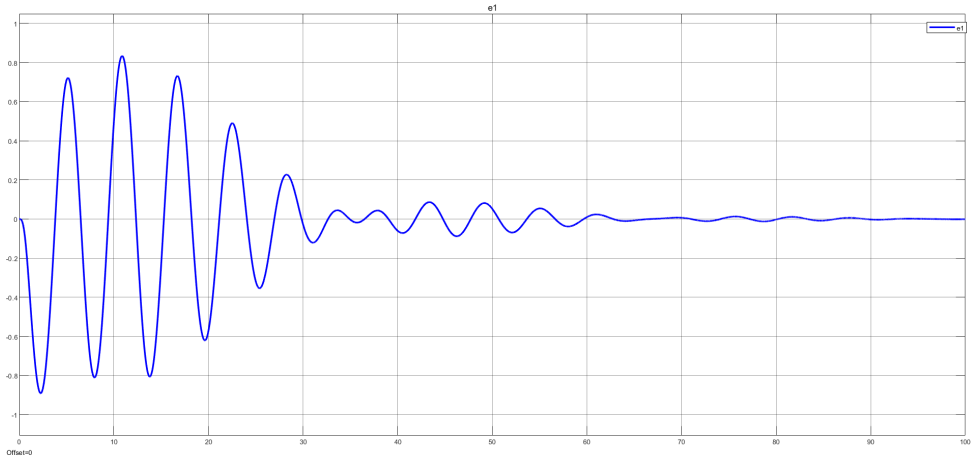


Figure 27: Tracking error e_1 under sinusoidal input

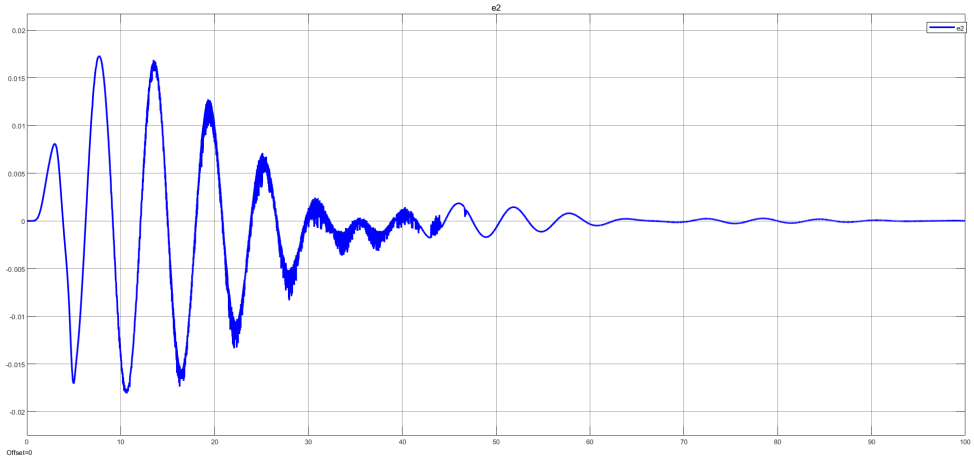


Figure 28: Auxiliary error e_2 under sinusoidal input

Conclusion (Question 3)

The simulation results confirm the theoretical analysis and demonstrate the effectiveness of the adaptive controller designed using Error Model 4 for a plant with relative degree three.

All errors — the tracking error e_1, the auxiliary error e_2, and the augmented error ϵ_1 — asymptotically converge to zero for both step and sinusoidal inputs. Although the settling time under the sinusoidal input was longer than that for the step input, the system successfully achieved convergence in both cases.

Furthermore, the adaptive parameters also converged, validating the correctness of the algebraic solution. The adaptation gains were selected to ensure fast and stable convergence.

The adaptation gain matrix for $\dot{\phi} \in \mathbb{R}^{6\times 1}$ is:

$$\Gamma_\phi = \begin{bmatrix} 3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 30 & 0 \\ 0 & 0 & 0 & 0 & 0 & 60 \end{bmatrix}$$

The scalar adaptation gain for $\dot{\psi}$ was:

$$\Gamma_\psi = 280$$

These selections allowed the adaptive system to maintain performance and stability, even with a non-SPR reference model, by properly constructing a stable augmented error structure.